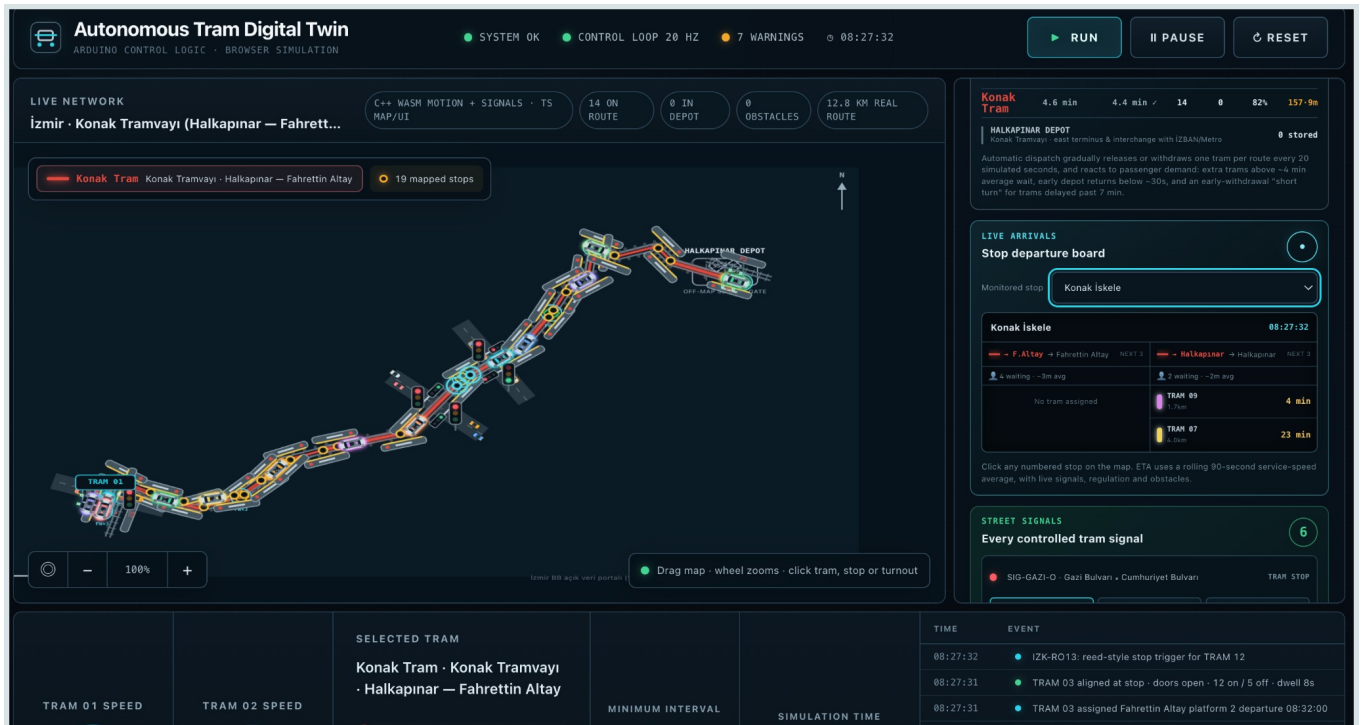


Autonomous Tram Digital Twin

Visual engineering brief for project discussion · September 2026

This prototype already combines vehicle physics, network infrastructure, passengers, dispatching and energy management in one deterministic multi-tram simulation. The strongest current research direction is coordinated energy-aware dispatch, not autonomous driving alone.

3	66	56	8.6–10.6%
implemented routes	directed Nizhny segments	flywheel modules	simulated grid-energy saving



Implemented İzmir network running with live vehicles, terminal service, ETA board, events and energy control.

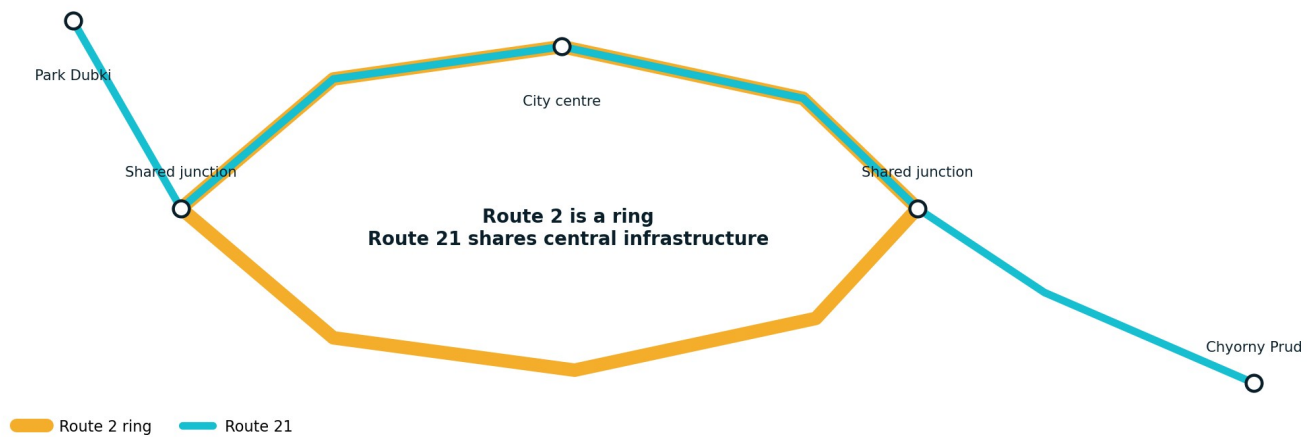
What is already operational

Vehicles and infrastructure	Service and passengers	Energy and research
Multi-tram motion · no overtaking · obstacles · road signals	Peak/off-peak plans · terminal departures · live ETA boards	Regeneration · section matching · flywheel storage
Switch locks · curve and turnout limits · depot portals	Passenger queues · capacity · boarding and alighting	Reactive and predictive dispatch · repeatable experiments
Door and stop authority in the C++ control core	Disturbance recovery · emergency vehicle release	Energy, peak power, SOC, rejected regen and losses

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Three Routes One Control Platform

Two interacting Nizhny services and one complete İzmir corridor



Operational topology of the Nizhny scenario. Route 2 is a ring; Route 21 crosses and shares central infrastructure.

Implemented network	Service geometry	Infrastructure represented
Nizhny Novgorod Route 2	17.44 km directed ring cycle	Gradients, central junctions, signals, turnouts and depot operations
Nizhny Novgorod Route 21	21.92 km Park Dubki–Chyorny Prud cycle	Shared Route 2 rails, merge conflicts and route-specific terminals
Integrated Nizhny 2 + 21	66 directed segments · 5 electrical sections	Shared ordering, independent headways and 38 flywheel modules
İzmir Konak Tram	12.8 km real corridor · 23.54 km two-way cycle · 19 stops	Six electrical sections, dual-track terminals and Halkapınar depot

Software responsibility is deliberately separated

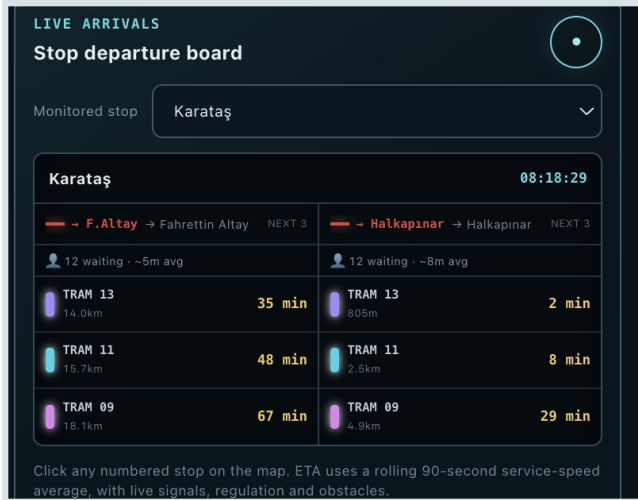
Layer	Technology	Authority
Deterministic control core	C++ compiled to WebAssembly ABI 11	Motion, braking, traction energy, movement caps, headway authority, signals, stops, doors, switches and depot state
Scenario and experiment layer	TypeScript	Track graph, service plans, passenger demand, ETA, logging and headless experiment orchestration
Human-facing interface	React and Canvas	Map rendering, live panels, statistics, manual controls, zoom and comparison mode

Architecture principle. The browser does not replace the controller. It visualizes and orchestrates the same stateful logic that can later be connected to Arduino telemetry or called from a MATLAB validation workflow.

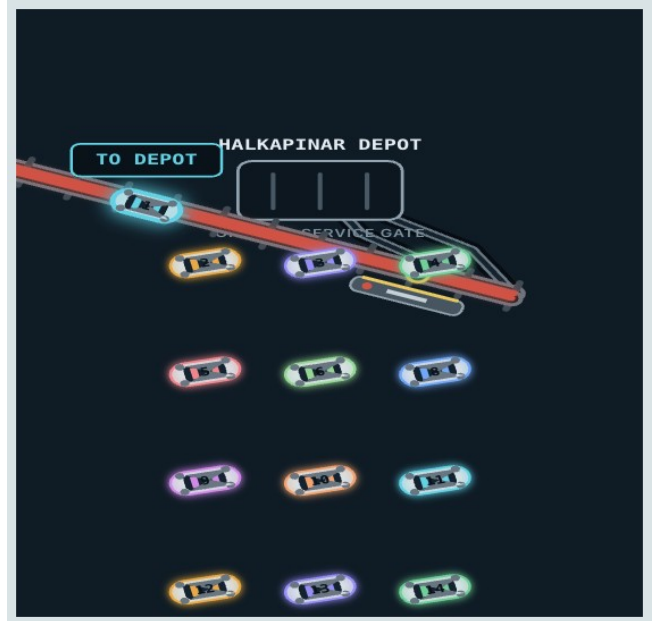
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Dispatch Passenger Service and Depot Logic

Closed-loop control for routine operation and disruptions



Two-direction stop board with rolling ETA and waiting passengers.



Halkapınar depot with controlled entry, storage and release.

Implemented operating decisions

Function	Trigger or rule	Implemented response
Terminal departure	Planned departure plus temporal headway to the preceding tram	On-time window ± 60 s; following tram is not released below 85% of planned headway
In-service anti-bunching	Live gap to the leading tram becomes too short	Speed authority is reduced without overriding obstacles, signals, doors or switch safety
Emergency extra tram	Estimated average wait > 10 min continuously for 5 min	One extra vehicle can be authorized for 20 min; 15 min cooldown prevents oscillation
Depot movement	Service plan requests entry or additional capacity	Ingress/egress states and rail-occupancy checks prevent instant appearance on the line
İzmir terminal capacity	Two terminal berths are occupied	A third tram waits at terminal entry instead of overlapping a platform vehicle

Passenger and door safeguards

- Time-of-day demand and stop importance create passenger queues; overflow remains when the 110-passenger capacity is reached.
- Boarding and alighting through three doors determine an 8–45 s dwell time rather than a fixed stop delay.
- Doors open only during valid station dwell, in passenger service and below 0.15 m/s; departure is blocked until boarding is complete.

Regularity metric. Headway CV = standard deviation of observed headways ÷ mean headway. A value of 0 is perfectly regular; 0.10 means the typical variation is about 10% of the mean interval.

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Physics Speed Profiles and Elevation

Passenger mass, smooth braking and infrastructure-dependent speed authority

Current reference tram parameters

Parameter	Current value	Role in simulation
Empty vehicle mass	27.5 t	Passenger mass is added dynamically to acceleration, braking and energy demand
Traction power limit	240 kW	Caps mechanical traction demand
Maximum acceleration	1.15 m/s ²	Applied through jerk-limited motion control
Service / emergency braking	1.25 / 2.40 m/s ²	Smooth station approach plus separate obstacle emergency authority
Jerk limit	0.90 m/s ³	Prevents an instantaneous change in acceleration or braking force
Auxiliary electrical load	4.5 kW	Represents continuous non-traction consumers
Regeneration conversion / receptivity	86% / 82%	Limits how much wheel-braking energy becomes accepted electrical energy

Forces and local speed restrictions

Model element	What is represented	Visible operational effect
Rolling and aerodynamic resistance	$A + Bv + Cv^2$ resistance model	Natural speed loss during coasting and higher demand at speed
Passenger loading	Current passenger count contributes to total mass	Loaded trams accelerate and brake differently and consume more energy
Curves	Segment-level curve speed profile	Tram reduces speed before the curve instead of braking at the curve point
Branch turnouts	15 km/h turnout authority	Safe low-speed traversal of switches and junction branches
Station approach	Jerk-limited target-speed trajectory	Smooth braking aligned with the platform rather than the map marker

Nizhny Novgorod elevation translated into energy

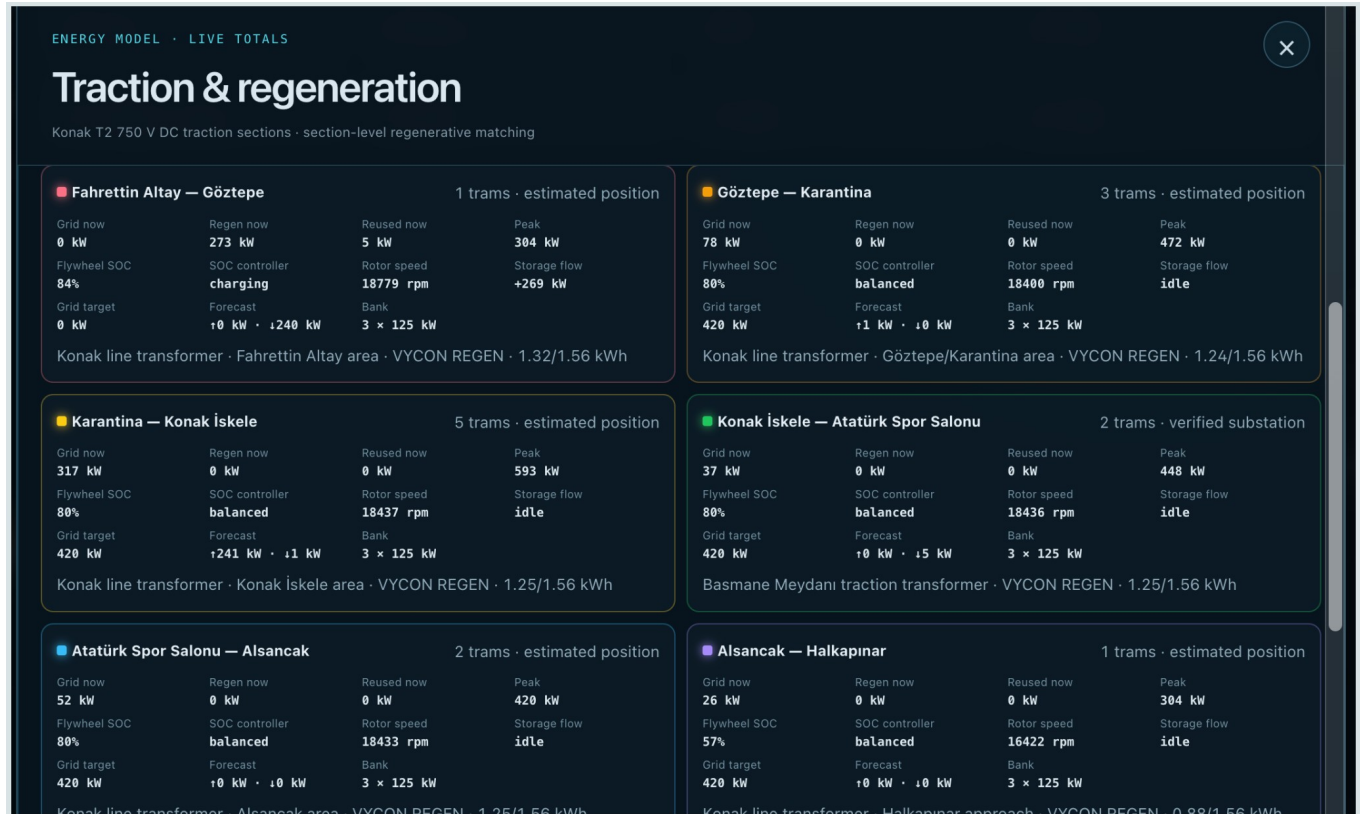
8	102 m	7.64 kWh	≈5.39 kWh
measured elevation differences	total descending elevation	calculated mgh per 27.5 t tram	theoretical accepted portion

The 5.39 kWh figure applies 86% regenerative conversion and 82% section receptivity to the dimensionally consistent 7.64 kWh result. An earlier project presentation stated 7.84 kWh; the difference is now documented rather than silently reproduced. Actual reuse is lower whenever no accelerating tram or flywheel capacity is available. Route 21 inherits the profile on shared track; its private branch is currently level.

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Flywheel Storage and Predictive Coordination

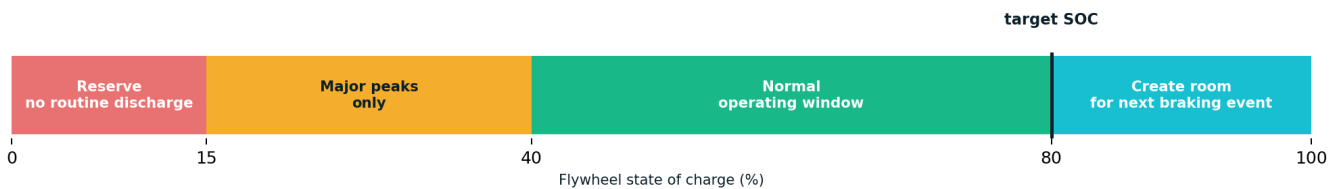
Real industrial reference hardware represented at electrical-section level



Live İzmir electrical sections. Each panel exposes grid demand, regeneration, reuse, SOC, rotor speed, storage flow, forecast and controller state.

Flywheel reference and installed simulation capacity

Item	Reference / implemented value	Meaning
Module	VYCON REGEN · 125 kW · 1,875 kW·s · 10,000–20,000 rpm	A real commercially deployed flywheel product used as the model reference
Nizhny network	38 modules across 5 sections	Section banks absorb braking energy and support local traction demand
İzmir network	18 modules across 6 sections	Three modules per simulated section
Energy priority	Direct tram-to-tram reuse → storage → grid / rejection	Avoids unnecessary conversion through storage when a local tram can consume regeneration



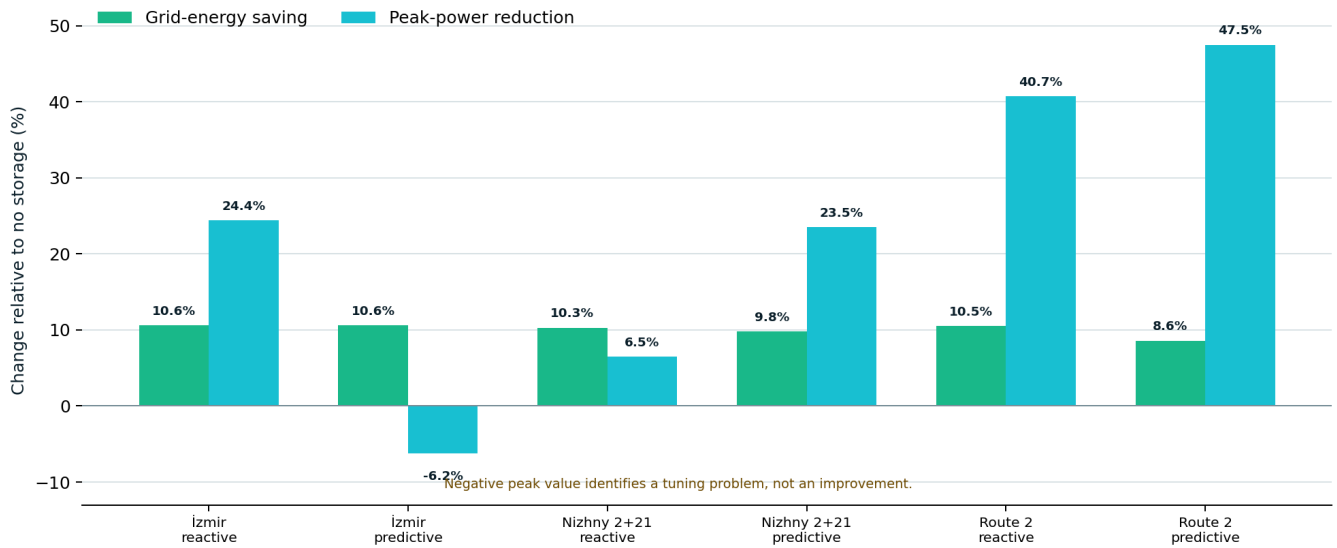
Implemented SOC policy. The controller normally oscillates around 80%, preserving both emergency reserve and braking headroom.

Energy-aware dispatch. The predictive controller estimates upcoming traction and braking inside each section. It avoids suppressing acceleration when another tram is already regenerating and can temporarily soften acceleration when forecast demand would create an unserved grid peak.

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Benchmarks Validation and Next Steps

Reproducible 10-minute experiments starting at 08:00



Change relative to the identical no-storage run. Energy values are normalized by vehicle-kilometre.

Scenario	Grid saving	Peak reduction	Interpretation
İzmir · reactive	10.6%	24.4%	Good baseline storage behavior with identical vehicle motion
İzmir · predictive	10.6%	-6.2%	Energy retained, but peak logic requires tuning
Nizhny 2+21 · reactive	10.3%	6.5%	Shared-network storage baseline
Nizhny 2+21 · predictive	9.8%	23.5%	120 dispatch interventions; substantially better peak shaving
Nizhny Route 2 · predictive	8.6%	47.5%	Strongest peak reduction in the current benchmark set

What these numbers do and do not prove

- They are deterministic simulator comparisons, not measured claims about savings achieved by either real operator.
- İzmir vehicle parameters still require Hyundai Rotem fleet calibration; cable resistance, voltage sag, detailed DC load flow and exact converter maps remain future work.
- The negative İzmir predictive peak result is retained because it reveals a real controller-tuning problem rather than hiding an inconvenient outcome.

Planned MATLAB and hardware validation

Next step	Purpose
Import prototype logs for speed, PWM, power, braking and position	Create a physical measurement baseline
Identify resistance, traction and braking parameters in MATLAB / Simulink	Calibrate vehicle and track physics
Build 600/750 V DC power-flow and flywheel converter reference models	Validate voltage, cable losses and storage behavior
Run identical JSON scenarios in MATLAB and C++/WASM	Quantify model error and reproducibility
Connect Arduino telemetry through hardware in the loop	Close the physical-digital control loop

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